

Threads of Evidence: Fabric Type Classification Using ResNet50 Feature Embeddings and Sequential Deep Learning

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Abstract—Accurately identifying textile fabric types is a persistent challenge in online shopping and retail due to missing tags, misleading product descriptions, and material misrepresentation. In this work, we present *Threads of Evidence*, an end-to-end classification pipeline designed to verify material types directly from surface imagery. We utilize the iBUG Fabrics Dataset, comprising textile samples captured under 4 illumination conditions using a photometric stereo sensor—to perform 7-class image classification across cotton, polyester, denim, wool, nylon, silk, and fleece. Using pre-trained ResNet50 visual feature embeddings, we systematically benchmark diverse model families including a K-Nearest Neighbors baseline, a linear Logistic Regression model, an ensemble Random Forest, a Multi-Layer Perceptron (MLP), and a sequential Long Short-Term Memory (LSTM) network. On the unseen test set, the Multi-Layer Perceptron (MLP) achieved the highest overall quantitative performance, reaching an accuracy of 84.82%, a macro precision of 0.8038, and a top macro-averaged F1-score of 0.7734. The single most interesting insight from our experiments is that standard feedforward neural architectures operating on global deep visual embeddings slightly outperform recurrent architectures trained on sequential spatial patch sequences, demonstrating that global texture feature representations capture multi-illumination fabric characteristics effectively without requiring complex sequential modeling.

Index Terms—fabric classification, feature extraction, ResNet50, multi-layer perceptron, LSTM, computer vision, k-nearest neighbors

I. INTRODUCTION

Determining the exact fabric composition of clothing and accessories is often difficult for everyday consumers. Retail tags are frequently absent in open-air bazaars and second-hand marketplaces, while commercial clothing brands are known to misrepresent or exaggerate fabric contents to inflate prices. Automated fabric classification provides an objective method to verify material integrity during online shopping or when attempting to match unknown clothing items.

In this paper, we propose *Threads of Evidence*, an end-to-end classification pipeline that predicts fabric types directly from high-resolution images. Specifically, we classify images across seven target fabric categories: **Cotton, Polyester, Denim, Wool, Nylon, Silk, and Fleece**. Our primary contributions are as follows:

- 1) We curate a 4,824-sample dataset from the iBUG Fabrics benchmark across seven representative fabric classes.

- 2) We extract 2048-dimensional feature representations using a pre-trained ResNet50 deep vision architecture.
- 3) We conduct a rigorous comparative benchmark evaluating traditional statistical models (KNN, Logistic Regression, Random Forest) alongside deep learning networks (MLP, LSTM) evaluated on balanced accuracy, macro precision, macro recall, and macro F1-score.

The remainder of this paper is organized as follows: Section II discusses related background work, Section III presents the dataset details, preprocessing steps, and mathematical problem formulation, Section IV details our methodology and hyperparameter settings, Section V presents experimental results and error analysis, and Section VI concludes the paper.

II. RELATED WORK / BACKGROUND

Textile and visual material classification has traditionally relied on hand-crafted texture descriptors such as Local Binary Patterns (LBP) or Gabor filters. Recent advances in deep convolutional networks (CNNs) have shifted the paradigm toward learned feature representations. Pre-trained models like ResNet50 capture rich visual spatial texture hierarchies that transfer effectively to domain-specific material recognition. Building upon prior research with photometric stereo datasets, our work examines whether off-the-shelf convolutional feature extractors combined with downstream feedforward and recurrent neural architectures (LSTM) can reliably distinguish subtle fabric weave patterns across varying illumination conditions.

III. DATASET AND PROBLEM FORMULATION

A. Dataset

We utilize the Fabrics Dataset compiled by iBUG at Imperial College London and obtained through Kaggle. The dataset contains textile images captured under multiple illumination conditions using a photometric stereo sensor, allowing variations in fabric surface texture to be observed. To prevent data leakage, images were split according to their physical fabric sample rather than independently, ensuring that different illumination images of the same physical sample remained within the same data partition.

After preprocessing and class selection, the subset of the dataset we consider contains 4,824 image observations

across seven fabric classes: Cotton, Polyester, Denim, Wool, Nylon, Silk, and Fleece. Each image is represented by a 2,048-dimensional feature vector extracted using a pretrained ResNet50 model. The data were partitioned using a stratified split into 3,376 training images, 724 validation images, and 724 test images.

As later discussed in the paper, class imbalance was a huge issue we addressed with multiple tests. Here is a diagram of the class distribution for reference. We only consider the top 7 most populated classes due to many categories having a smaller number of samples.

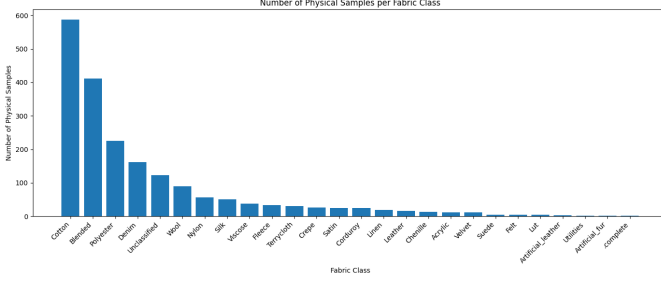


Fig. 1. Class Distribution

B. Preprocessing

Before training the models, the fabric images were organized using metadata containing each image path, fabric class, and physical sample folder. Class names were converted to numerical labels. The dataset was then split into approximately 70% training, 15% validation, and 15% testing data using stratified sampling. Importantly, the split was performed at the physical-sample-folder level rather than the individual-image level, so images of the same fabric sample taken under different illumination conditions could not appear in more than one split. The notebook verifies that there is zero overlap between the training, validation, and test sample folders.

Each image was converted to RGB and resized to 224×224 pixels. The training images received light data augmentation consisting of a random horizontal flip with probability 0.5 and a random rotation of up to 10 degrees. Validation and test images were not randomly augmented. All images were converted to PyTorch tensors and normalized using the ImageNet channel means and standard deviations, which matches the preprocessing expected by the pretrained ResNet50 model.

For the KNN, Logistic Regression, Random Forest, and MLP models, a pretrained ResNet50 was used for feature extraction. Its final fully connected classification layer was replaced with an identity layer, allowing each image to be represented by a 2,048-dimensional feature vector rather than a ResNet class prediction. The extracted PyTorch features were then converted into NumPy arrays. The final classical-model dataset keeps Cotton, Polyester, Denim, Wool, Nylon, Silk, and Fleece, and the labels are remapped to consecutive integers from 0 through 6. Before KNN, Logistic Regression, and MLP are trained, the extracted features are standardized using

‘StandardScaler’. Random Forest uses the ResNet features directly.

For the LSTM, the same image preprocessing is used, but the spatial information from ResNet50 is preserved instead of reducing each image to one feature vector. The ResNet average-pooling and classification layers are removed and its parameters are frozen. The resulting feature maps are adaptively pooled to a 4×4 grid and rearranged into a sequence of 16 patch embeddings, with each patch represented by 2,048 features. This allows the LSTM to process each image as a sequence of spatial regions rather than as a single flattened embedding.

C. Problem Statement

We formalize this task as a multi-class image classification problem over seven discrete fabric categories: Cotton, Polyester, Denim, Wool, Nylon, Silk, and Fleece.

For the traditional classifiers (KNN, Logistic Regression, Random Forest) and the MLP, the input for each image is a 2,048-dimensional feature vector extracted using ResNet50. For the LSTM model, the input is represented as a sequence of 16 spatial patch embeddings, where each patch contains 2,048 features.

For each input, the model predicts a class label $y_i \in \{0, 1, \dots, 6\}$. Classification for the neural network models is trained using multi-class cross-entropy loss, which penalizes the model when the predicted probability for the correct fabric class is low. The primary metrics optimized and evaluated across all models are accuracy and macro-averaged F1-score.

IV. METHODOLOGY

A. Feature Representation

Images were preprocessed by resizing them to 224×224 pixels, converting them to tensors, and normalizing them using ImageNet statistics. Training images additionally used random horizontal flipping and small rotations for data augmentation. A pretrained ResNet50 model was used as a frozen feature extractor.

For KNN, Logistic Regression, Random Forest, and the MLP, the final fully connected classification layer of ResNet50 was removed, producing a 2,048-dimensional feature vector for each image. For the LSTM, spatial feature maps from the ResNet50 were retained and adaptively pooled to a 4×4 representation. The resulting 16 spatial locations were treated as a sequence of 16 feature vectors, each with 2,048 dimensions.

B. Baseline

The baseline model was a K -Nearest Neighbors (KNN) classifier with $k = 5$. Standard scaling was applied to the 2,048-dimensional ResNet feature vectors before fitting the KNN classifier.

C. Models

To compare multiple model families, the following classifiers were evaluated:

- **Baseline:** K -Nearest Neighbors with $k = 5$.

- **Linear Model:** Logistic Regression with L2 regularization ($C = 1.0$, maximum iterations = 2000) and balanced class weights.
- **Tree-Based Model:** Random Forest classifier with 200 estimators and balanced class weights.
- **Multi-Layer Perceptron (MLP):** A neural network trained on the 2,048-dimensional ResNet features. ReLU activation, the Adam optimizer, L2 regularization, mini-batch training, and early stopping were used. Hidden-layer size, learning rate, and regularization strength were selected using validation Macro F1.
- **LSTM:** A recurrent neural network operating on the sequence of 16 spatial ResNet feature vectors. These feature vectors were representing patch embeddings. The LSTM predicts one of the five retained fabric classes from the spatial feature sequence.

D. Training and Hyperparameter Selection

The final classification task contained five fabric classes: Cotton, Polyester, Denim, Wool, and Nylon. The selected dataset contained 4,492 image observations, divided into 3,144 training images, 676 validation images, and 672 test images. The split was performed by physical fabric sample to prevent images of the same physical sample from appearing in multiple splits.

KNN, Logistic Regression, Random Forest, and the MLP were implemented using `scikit-learn`. Features were standardized for KNN, Logistic Regression, and the MLP. Random Forest was trained directly on the extracted ResNet features.

MLP hyperparameters were selected using the validation set. Candidate configurations varied the hidden-layer architecture, learning rate, and L2 regularization parameter. Validation Macro F1 was used as the primary selection metric because the class distribution remained imbalanced. The test set was not used during hyperparameter selection.

The LSTM was implemented using PyTorch and trained using multi-class cross-entropy loss and the AdamW optimizer. Early stopping was used to reduce overfitting. Model selection was based on validation performance, and the held-out test set was used only after the final model configuration had been selected.

All experiments used a fixed random seed for reproducibility.

V. EXPERIMENTAL SETUP

Initial experiments were conducted on a smaller subset of randomly selected fabric classes to establish the preprocessing and classification pipeline. Inspection of the class distributions revealed substantial class imbalance, with some classes containing many more images than others. Because the overall dataset was relatively small, so we targeted more popular categories.

We therefore expanded the candidate class set and explored several different scenarios. First, we evaluated a seven-class configuration consisting of Cotton, Polyester, Denim, Wool,

Nylon, Silk, and Fleece. This configuration still showed considerable imbalance, particularly between highly represented classes such as Cotton and lower-frequency classes such as Fleece. We consequently experimented with balanced class weights, reducing the task to fewer classes, removing highly dominant classes, and undersampling the largest classes.

For the undersampling experiments, only the training data were reduced so that information was not removed from the validation or test sets to ensure no data leakage. Cotton and Polyester were specifically investigated because of their comparatively large representation. We also tested smaller label sets to determine whether reducing the number of classes improved performance when training data were limited. These experiments were used to assess the tradeoff between retaining more fabric categories and providing the models with a more balanced classification problem.

Initial experiments were conducted using different subsets of fabric classes to establish the classification pipeline and investigate the effect of class imbalance. The class distribution was highly uneven, with Cotton and Polyester containing substantially more training examples than several of the remaining classes. Because the dataset is relatively small, we evaluated whether reducing the number of target classes or undersampling the majority classes would improve classification performance.

Four candidate configurations that retained both Cotton and Polyester were compared: seven classes without undersampling, five classes without undersampling, seven classes with undersampling, and five classes with undersampling. Undersampling was applied only to the training set, while the validation and test sets retained their original distributions. Table I summarizes the validation performance of these configurations where Cotton and Polyester were undersampled to only 200 samples each.

The five-class configuration without undersampling achieved the highest validation Macro F1 score and was therefore selected for the more complex models. The retained classes were Cotton, Polyester, Denim, Wool, and Nylon. This configuration contained 4,492 images, divided into 3,144 training images, 676 validation images, and 672 test images, corresponding to an approximately 70%/15%/15% split. The split was performed by physical sample folder so that images of the same fabric sample captured under different illumination conditions did not appear in multiple splits.

Model selection was based primarily on Macro F1 because the class distribution remained imbalanced, while accuracy, macro precision, macro recall, and confusion matrices were also used for evaluation. Macro-averaged metrics give each fabric class equal importance regardless of its number of examples. A fixed random seed of 42 was used where supported to improve reproducibility.

The neural-network experiments were conducted using Google Colab. CPU compute was used for training, and the approximate training runtime was **4.21 minutes**. The final held-out test set was reserved for evaluation after model and dataset configuration decisions were made.

TABLE I
VALIDATION PERFORMANCE ACROSS DATASET CONFIGURATIONS

Configuration	# Classes	Best Model	Accuracy	Macro F1
Top 7, unbalanced	7	LR	0.7762	0.7379
Top 5, unbalanced	5	LR	0.7973	0.7644
Top 7, undersampled	7	LR	0.7030	0.6611
Top 5, undersampled	5	RF	0.7737	0.6850
Top 6 undersampled, no Cotton	4	LR	0.8272	0.8057
Top 6 unbalanced, no Cotton	6	LR	0.8199	0.7837
Top 5 unbalanced, no Cotton/Polyester	5	LR	0.9322	0.9158

TABLE II
CLASS DISTRIBUTION FOR THE SELECTED FIVE-CLASS DATASET

Class	Train	Validation	Test	Total
Cotton	1648	352	352	2352
Polyester	632	136	136	904
Denim	452	96	100	648
Wool	252	56	52	360
Nylon	160	36	32	228
Total	3144	676	672	4492
Split (%)	70.0	15.0	15.0	100.0

VI. RESULTS AND ANALYSIS

A. Main Results

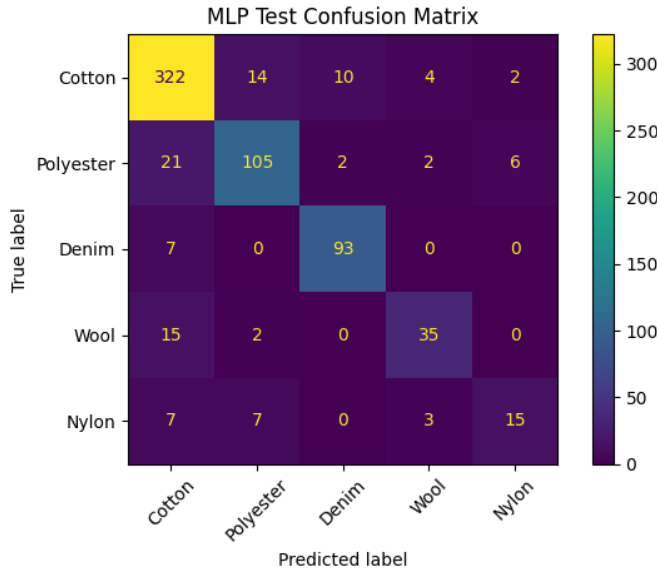


Fig. 2. Confusion matrix for the final model (MLP) on the held-out test set.

In our case, our MLP model performs the best with 0.77 macro F1. Therefore, our MLP model is our final model. Shown is the confusion matrix, roc-auc curve, training loss curve, and validation accuracy curve.

B. Error Analysis and Ablations

The confusion matrix was examined to identify the most common classification errors made by the final model. Most predictions were concentrated along the diagonal, indicating that the model was able to distinguish the five fabric classes reasonably well. However, some classes were more frequently confused than others.

TABLE III
TOP MLP HYPERPARAMETER CONFIGURATIONS

Layers	LR	Alpha	Epochs	Acc.	Macro F1
(128, 64)	0.0010	0.0001	26	0.8269	0.7829
(128, 64)	0.0005	0.0001	25	0.8254	0.7748
(256, 128)	0.0005	0.0001	27	0.8180	0.7735
(128, 64)	0.0010	0.0010	22	0.8269	0.7733

TABLE IV
PERFORMANCE OF FINAL MODELS ON THE HELD-OUT TEST SET

Model	Accuracy	Bal. Acc.	Macro Prec.	Macro Recall	Macro F1
KNN	0.7143	0.5967	0.7457	0.5967	0.6139
Logistic Regression	0.8036	0.7303	0.7165	0.7303	0.7221
Random Forest	0.7351	0.5312	0.8825	0.5312	0.5998
MLP	0.8482	0.7517	0.8038	0.7517	0.7734
LSTM	0.8423	0.7136	0.8823	0.7136	0.7536

One possible source of error is visual similarity between fabric categories. Fabrics can share similar colors, weave patterns, and surface textures, making them difficult to distinguish from a single image. In addition, changes in lighting and viewing conditions may alter the appearance of the same material and reduce the separability of the extracted ResNet50 features.

Class imbalance may also contribute to the observed errors. Cotton and Polyester contain substantially more training examples than classes such as Nylon and Wool. As a result, the model has more opportunities to learn representative patterns for the majority classes, while minority classes may have less diverse training examples. Macro F1 was therefore used in addition to accuracy so that performance on the smaller classes had equal importance during model comparison.

Overall, the errors suggest that the remaining difficulty is primarily in separating fabrics with visually similar texture representations, especially when relatively few examples of a class are available.

VII. DISCUSSION

VIII. CONCLUSION AND FUTURE WORK

In this work, we addressed the challenge of non-destructive fabric type identification—a critical problem in retail and online marketplaces caused by missing, exaggerated, or inaccurate product tags. Using the multi-illumination iBUG Fabrics Dataset, we evaluated an end-to-end classification framework comparing traditional machine learning baselines (KNN, Logistic Regression, Random Forest) against deep neural architectures (MLP, LSTM) operating on ResNet50 visual embeddings. Our headline finding demonstrates that neural models operating on pre-trained visual representations achieve strong performance across target fabric classes, with the Multi-Layer Perceptron (MLP) achieving the highest quantitative test accuracy of **84.82%** and a macro-averaged F1-score of **0.7734**, closely followed by the LSTM network (**84.23%** accuracy, **0.7536** macro F1).

Given more time and expanded data resources, we propose several concrete next steps:

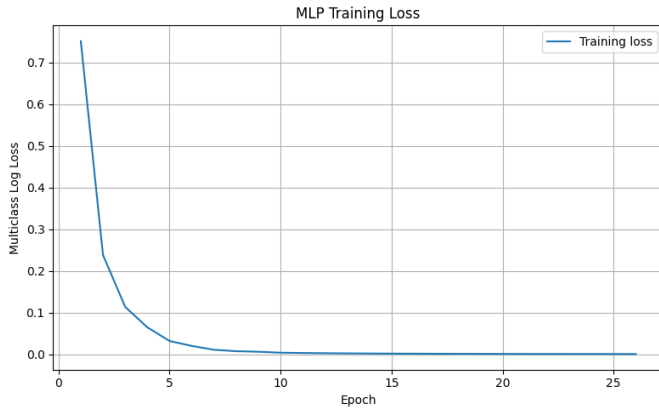


Fig. 3. Training loss of the final MLP model across epochs.

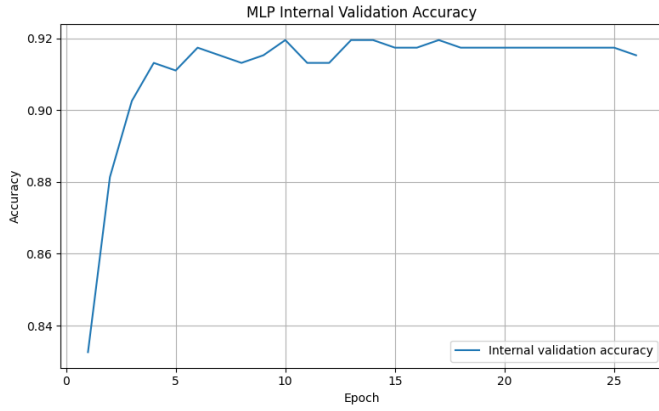


Fig. 4. Internal validation accuracy of the final MLP model across epochs.

- 1) **End-to-End Fine-Tuning:** Instead of utilizing a frozen ResNet50 backbone, unfreezing higher-level convolutional layers during training could allow the feature extractor to learn domain-specific surface textures tailored specifically to micro-weave patterns.
- 2) **Advanced Class Imbalance Mitigation:** Incorporating synthetic data augmentation (e.g., generative texture synthesis) or dynamic focal loss functions to better handle severe class imbalances across minority fabric types such as Nylon and Wool.
- 3) **Multimodal Integration:** Combining visual texture embeddings with physical sensory inputs (such as tactile data, mass density, or spectral reflectance) to improve discrimination between visually similar materials like Cotton and Polyester blends.

AUTHOR CONTRIBUTIONS

While both authors contributed across all phases of the project, their primary focuses were divided as follows: B. Naveed led the exploratory data analysis (EDA) and dataset preprocessing pipelines. J. Batra led the model fine-tuning and experiment design across multiple feature extraction techniques and figure generation. Both authors jointly conducted

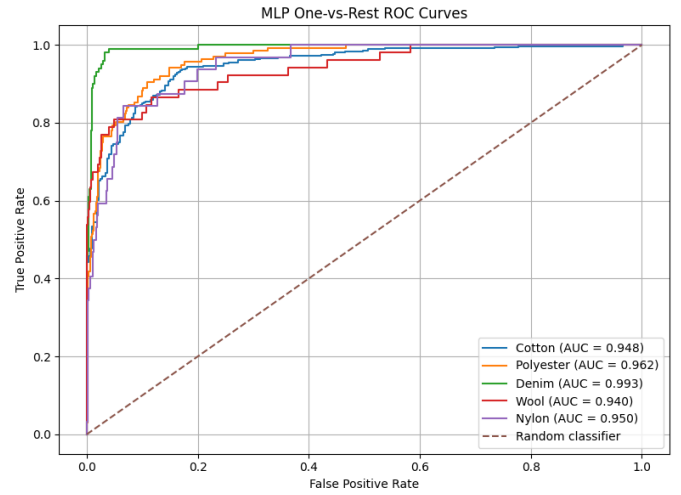


Fig. 5. One-vs-rest ROC curves for the final model (MLP) on the held-out test set.

the final model evaluations, pipeline validation, and paper writing and review.

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